

Compromised in Silence: Analysing Zero-Click Exploits and Acoustic Surveillance Threats

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Abstract—The proliferation of always-connected smart devices has given rise to a class of attacks that require no user interaction and leave no visible trace. This research examines two such threats: zero-click attacks and acoustic covert communication. Zero-click attacks exploit vulnerabilities in how operating systems and applications automatically process incoming data — such as images, audio, and calendar invitations — enabling remote device compromise before a user even sees a notification. Acoustic covert communication exploits the ubiquitous microphones and speakers in smart devices to embed inaudible ultrasonic beacons that silently profile and track users across platforms and locations, without requesting any permissions. Together, these threats represent a fundamental shift in the attack landscape: the user is no longer the weakest link, because the user is no longer part of the equation. This research analyses the technical mechanisms behind both threat categories, presents real-world case studies, and reviews defensive strategies including Zero-Trust Architecture (ZTA), high-frequency audio filtering, and endpoint protection.

Index Terms—Zero-Click Attacks, Acoustic Covert Communication, Cross-Device Tracking, Zero-Trust Architecture, Privacy.

I. INTRODUCTION

Modern cyberattacks no longer require a user to make a mistake. Zero-click exploits and acoustic covert tracking compromise devices and collect personal data silently, triggered by a received message, a background audio file, or an inaudible beacon embedded in a TV advertisement.

A common misconception is that only important people — politicians, executives, or journalists — are worth targeting. In reality, ordinary users are the primary target at scale. Advertisers and data brokers profit from tracking everyday browsing habits, location history, and media consumption. Criminals harvest banking credentials and personal data from regular users in bulk. Even surveillance tools like Pegasus have been found on the phones of ordinary individuals connected to persons of interest. The value is not in who you are — it is in the data your device continuously generates.

Privacy is not a privilege reserved for the powerful. It is a fundamental right that enables personal freedom, autonomy, and dignity. When our devices are silently compromised, it is not just data that is lost — it is the ability to speak freely, move without being tracked, and make decisions without being profiled and manipulated. An ordinary person's medical searches, financial struggles, or private conversations are deeply personal, and their exposure can lead to real consequences: targeted scams, insurance discrimination, emotional manipulation, or simply the loss of control over one's own narrative. Protecting ourselves from silent attacks is therefore not a technical luxury, it is a matter of personal security in the most human sense.

The purpose of this research is to make these threats understandable and to demonstrate that silent attacks are not an abstract concern. Understanding how they work and what defences exist is the first step toward meaningful protection, regardless of whether the reader considers themselves a high-value target or not. Beyond the two threats covered here, silent attacks also extend to IoT devices, telecommunications exploits, and side-channel techniques, all of which are acknowledged as part of the broader threat landscape but fall outside the scope of this research. The following sections focus specifically on zero-click attacks and acoustic covert communication.

II. ZERO-CLICK ATTACKS

A. Introduction to zero-click attacks

To define A zero-click attack, so we can all be set on the same level of understanding of what a "zero-click" means, we can look at the definition of Younis A which he state "a zero-click attack is a remote code execution (RCE) paradigm where a device is compromised without any user interaction." [2], this means that once the device have been compromised the normal average person could not tell anything has been changed in the device and will use it normally like he would, even the tech savvy person may not have any idea of what is going on since it is a process in the background of the OS which is very hard to find and could be even be changed to look like a normal process that the OS runs. Since the exploit allows the intruder to execute code unnoticeably, the intruder could inject the device with spyware, malware, and all kinds of viruses that could harm the end user with an irreversible effect.

Pegasus is one of the most controversial zero-click attacks in recent years. The origin of Pegasus is believed to be from NSO. It could be deployed on devices running Android or iOS, which is practically every phone on the market. How is it deployed when the phone connects to the internet Pegasus will send a "bad code" to the phone by different methods. Once the device on the receiving end gets the message, the code will run, and the phone is now an active 24-hour surveillance tool in the hands of the NSO. It will give the NSO unrestricted access to the phone so they can get any data they want. And this is not being deployed into someone whom the NSO is targeting; it is a software that was tracked to be deployed into a normal person's phone. The Citizen Lab-University of Toronto, and Amnesty's lab have successfully tracked and reported on an iPhone on which Pegasus was deployed. They noticed multiple ways that the phone could be compromised by Pegasus, a total of 65 techniques of 13 different tactics could be used to deploy the software [2], [3].

TABLE I
SET OF ATTACK MAPPINGS (ZERO-ATTACKS PROFILE)

Subject	Id	Tool
Initial Access (Initial)		
Initial Access (Initial)	201401	1. The user is a targeted user who receives a text message with a link.
Initial Access (Initial)	201402	2. The user is a targeted user who receives a text message with a link.
Initial Access (Initial)	201403	3. The user is a targeted user who receives a text message with a link.
Execution (Initial)		
Execution (Initial)	201404	1. The user is a targeted user who receives a text message with a link.
Execution (Initial)	201405	2. The user is a targeted user who receives a text message with a link.
Execution (Initial)	201406	3. The user is a targeted user who receives a text message with a link.
Privilege Escalation (Initial)		
Privilege Escalation (Initial)	201407	1. The user is a targeted user who receives a text message with a link.
Privilege Escalation (Initial)	201408	2. The user is a targeted user who receives a text message with a link.
Privilege Escalation (Initial)	201409	3. The user is a targeted user who receives a text message with a link.
Discovery (Initial)		
Discovery (Initial)	201410	1. The user is a targeted user who receives a text message with a link.
Discovery (Initial)	201411	2. The user is a targeted user who receives a text message with a link.
Discovery (Initial)	201412	3. The user is a targeted user who receives a text message with a link.
Collection (Initial)		
Collection (Initial)	201413	1. The user is a targeted user who receives a text message with a link.
Collection (Initial)	201414	2. The user is a targeted user who receives a text message with a link.
Collection (Initial)	201415	3. The user is a targeted user who receives a text message with a link.
Exfiltration (Initial)		
Exfiltration (Initial)	201416	1. The user is a targeted user who receives a text message with a link.
Exfiltration (Initial)	201417	2. The user is a targeted user who receives a text message with a link.
Exfiltration (Initial)	201418	3. The user is a targeted user who receives a text message with a link.

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Fig. 1. This is a mapping of the 63 ways [3]

Pegasus is not the only zero-click attack software, zero click attack could be deployed by hackers using the same methods as Pegasus has used. messages, PDF, an image, even a call or a miscall, are ways to activate the bad code being sent by the software. As discussed before, this gives the attacker full control of the device without any noticeable traces. So the attacker has access to private communications since end-to-end encryption is useless in this case, and the attacker has access to all files in the device, GPS location, sensitive credentials, incoming calls, which the attacker could record, microphone, and camera; all of these are in the hands of the attacker, The attacker could use AI agents to help mass surveillance of all the compromised devices and when a valuable info is shared the agent just send it to the attacker so he can use it to get what he wants. [3]

B. Case Studies: Zero-Click Attack Vectors and Targets

The nature of a zero-click attack is technically complex, resource-demanding, and expensive. So, it's not easy for a single hacker or a small group of hackers to deploy such software. Usually, nation-states and elite hacker groups deploy these kinds of software on high-value targets and organizations. [1].

Some high-profile examples include:

- i. **WhatsApp (Missed call):** Attackers in 2019 found a way to install spyware into the targeted phone by calling the target. Whether the target answers or not the sent packets are enough for the code to run [1], [8].

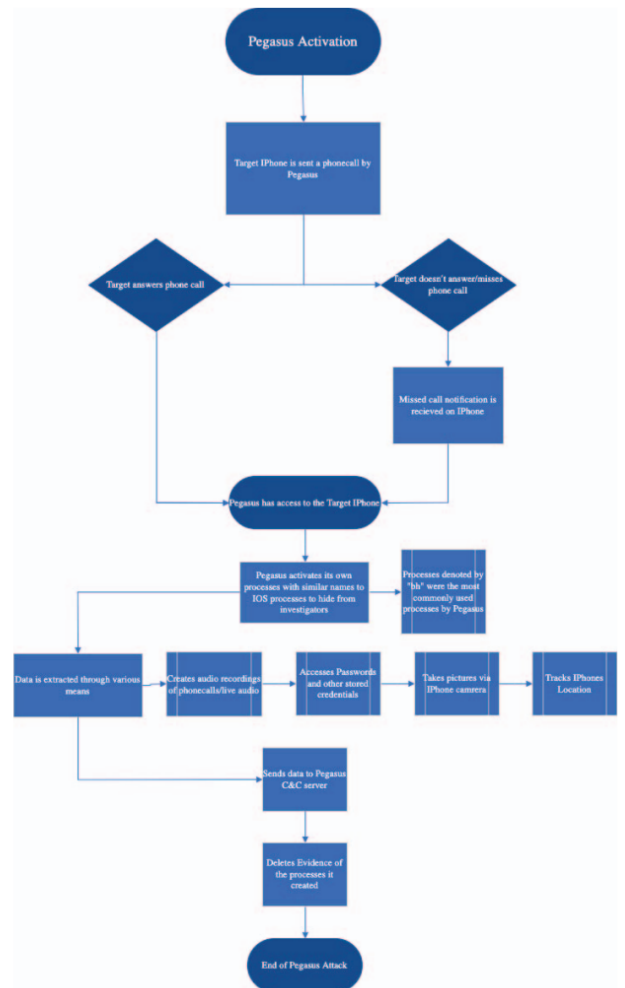


Fig. 2. This is a flow chart of how Pegasus works and other zero-click attacks [3].

- ii. **FORCEDENTRY (iMessage):** An exploit called FORCEDENTRY was found in 2021, which allowed the attacker to send a hidden PDF that has the bad code to be sent as a GIF, which Apple's iMessages engine tries to process it which activates the execution process [10].
- iii. **Outlook (Calendar reminders):** Outlook exploit in 2023, allowed attackers to send a calendar invitation to random people which was processed to set reminder sound, that allowed the attacker to steal the target password hashes [1], [9].
- iv. **Media processing vulnerabilities:** some operating systems like android is driven by an "always-on" nature, which include libraries that process media in a complex way, attackers target vulnerabilities in these media processing libraries. example of these are GIF and PDF which WhatsApp process and render a preview in the background for better user experience which will result in the execution of the bad code by triggering an overflow integer or buffer vulnerability in these media processing libraries. [1].

C. Zero-trust architecture: the modern defence framework

”Never trust, always verify” is the new way of thinking of security models. with the existence of zero-click attacks you always assume that a breach has happened or inevitably will happen which will be the foundational premise of (ZTA) Zero-Trust Architecture [1].

The current networking architecture relies on perimeter-based model, which grants trust to local device in the network to send and receive data from-to each other. with zero-click attacks this needs to change to user-based trust, since if 1 device in the old model is compromised the other all the networks will follow. So new model needs to be made following the ZTA to create a secure local network. [1].

The foundational NIST SP 800-207 standard, assume that we all and any data that the user receive is already in the danger zone, So all data needs to be Fully authenticated before execution. the reliance on physical or network location is not secure anymore. individual resources need to be Fully diagnosed before allowing it into the device which create resilient layer of defence. [1], [4].

A roadmap has already been structured by the CISA which will transform all security systems from the old models to ZTA models. and five main frameworks pillars will be set to help with this [1], [4]:

- i. **Identity:** Moving beyond simple passwords to robust Multi-Factor Authentication (MFA) and continuous risk-based identity verification.
- ii. **Devices:** Ensuring every device (corporate-owned or your own device) is known, managed, and verified for security health before accessing resources.
- iii. **Networks:** Transitioning from flat, open networks to micro-segmented environments where internal traffic is restricted.
- iv. **Applications and workloads:** Securing the actual software and cloud services where data is processed, ensuring they are isolated and verified.
- v. **Data:** The ultimate focus of Zero Trust; ensuring data is encrypted, categorised, and protected regardless of where it resides.

1) *Why ZTA is critical for zero-click defence:* the unique position of ZTA to mitigate the harmful impact of zero-click attacks. in ZTA the environment, the lateral movement that the attacker is trying to do by moving from infected device to high-value local or cloud server that is connected to the device is completely and effectively neutralized by using two mechanisms:

- i. **Continuous device health monitoring and revocation:** legacy models allow the authenticated device session to remain valid for an extensive amount of time, while on the other hand ZTA preforms a continuous validation process. so, if the device is infected in the old model, it could gain access to the server or device files, while in the ZTA the automatic ZTA agent detect if this new request by the device is an anomaly or not such as library addition or security changes or requiting to gain access to higher level files. this will result on the ZTA to cut off

all communication with device and revoke any access or token it had.

- ii. **Micro-segmentation and the “blast radius”:** The real threat of a zero-click attack is its ability to turn a single smartphone into a backdoor to a company’s network, but Zero Trust Architecture counters this by breaking the network down into separate, isolated areas. This means that even if a device is hacked, it immediately gets locked into its own restricted zone, effectively stopping hackers from scanning the network or moving laterally into the main database. Because any new asset request is blocked by default until it receives explicit clearance from the Policy Decision Point, the potential damage is completely limited to that single infected device, keeping the organization’s core assets safe. [1].

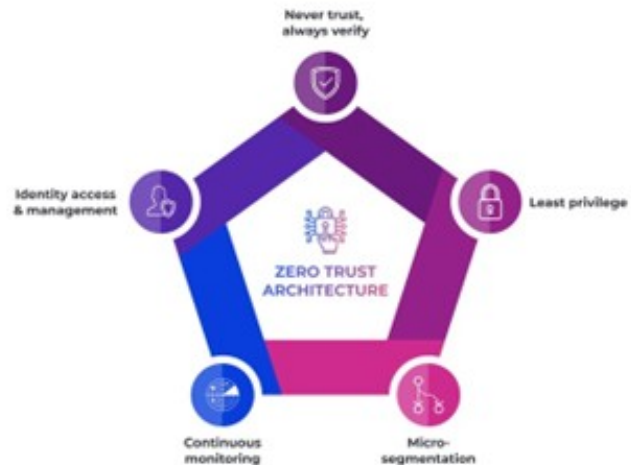


Fig. 3. five components of ZTA [3].

2) *AI-driven zero-trust architectures:* Integrating AI and machine learning into Zero Trust Architecture shifts network security away from rigid fixed rules and toward an active adaptive defense that can analyze user and device behavior in real time. Traditional Zero Trust systems are often fragile and struggle against sophisticated threats because they rely entirely on static policies set by administrators. By adding AI into the framework the system can continuously monitor behavior data to automatically adjust trust scores on the fly creating a highly flexible and proactive security posture that stays ahead of evolving risks and makes it significantly harder for attackers to succeed. [1].

D. Other Defensive Mechanisms Against Zero-Click Attacks

- 1) **Patch management:** Users can increase their security by regularly applying security patches and enforce automatic updates to software, firmware, and operating systems. This helps mitigate the risk of vulnerabilities that could be exploited in zero-click attacks and minimizes exposure to exploitable weaknesses [5].

- 2) **Virtualization:** Employing virtualization technologies, such as deploying virtual machines (VMs), Virtual Desktop Infrastructure (VDI) and virtual private servers (VPS) can provide a more secure temporary environment than the underlying host system to execute applications, services, and files. If malware compromises one application or service, the virtualization environment can be discarded periodically to prevent malware dwell time and reduce the risk of lateral movement and data exfiltration [5].
Annotation: Instead of a giant VM that looks like a second computer, as end user it may seem not practical to use VM whenever you want to open a PC, thus employing Micro-Virtualization is a sufficient solution, such that every time you open an email attachment or a browser tab, the system automatically creates a tiny and invisible VM just for that one task [6].
- 3) **Endpoint and Mobile protection:** Securing the devices themselves is the first line of defense. Organizations and individuals should deploy robust Endpoint Detection and Response (EDR) solutions alongside mobile threat detection tools and standard antivirus software. Rather than just looking for known viruses, these tools actively monitor the device for behavioral red flags—such as unusual memory spikes, unexpected background processes, unauthorized privilege escalations, or suspicious network connections—to catch malicious code the moment it tries to execute [5], [7].
- 4) **Vulnerability Scanning, Content Filtering, Disable Auto-Download in Messaging Apps:** Using vulnerability scanning software for all internal network and public facing infrastructure and ensuring that it is regularly updated with tests to find the most recently disclosed vulnerabilities and weak configurations. Utilise content email filtering tools to detect and block spam and more targeted phishing emails containing phishing links, or malicious content. Also, file format validation can be used to verify that a file's format strictly follows the expected format for a given file type. Furthermore, an essential more step is to disable "Auto-Download" in messaging apps such as WhatsApp. It prevents the system from automatically parsing the file as soon as it arrives. The file will not be opened unless the user manually chooses to do so. The user often notices suspicious messages or unknown senders and simply not open them, stopping the attack before it starts. Moreover, an attacker may send thousands of messages, targeting thousands of users. Auto-download ensures all those devices are infected instantly. Manual downloading makes the attack slow and ineffective at scale [4], [5], [7].
- 5) **Network security controls:** A strong defense requires keeping a close eye on the traffic flowing in and out of the network. Deploying advanced network controls—such as Intrusion Detection and Prevention Systems (IDS/IPS), Next-Generation Firewalls (NGFW), and Secure Web Gateways (SWG)—helps monitor traffic for the hidden payloads associated with zero-click attacks. Additionally, practicing strict network segmen-

tation using firewalls, VLANs, and access control lists is vital. Segmentation ensures that even if an attacker manages to silently compromise a single device, their access is quarantined, preventing them from moving freely through the rest of the network [5], [7].

Remark: What should do when suspect a zero-click attack? Immediately disconnect the device from networks, run security scans, check for unusual activity, update all software, and consult cybersecurity professionals for forensic analysis.

III. ACOUSTIC COVERT COMMUNICATION AND ZERO-PERMISSION ACOUSTIC CROSS-DEVICE TRACKING

A. Cross-device tracking

Companies today collect a massive amount of user data to improve customer experiences and boost financial profits through highly targeted advertising. Because advertising components are embedded across countless websites and TV streams, user behavior can be monitored on a grand scale. With everyone using mobile and wearable devices, there is a huge demand for cross-device tracking, which allows companies to connect your activities across different platforms—like your phone, laptop, tablet, and smart TV—creating a single, unified profile of your digital journey [11].

Usually, this tracking relies on explicit data, like a shared password used across different apps, or by comparing shared characteristics like IP addresses and locations. The ultimate goal is to figure out if the person on a specific computer is the same person using a specific mobile phone. This is incredibly valuable for marketers wanting to serve targeted ads regardless of the device you are using. While tech giants like Facebook and Google offer these services, they generally require you to be signed into their platforms [12].

This creates a serious privacy threat. These combined tracking profiles can hold deeply sensitive details about your location, health, personal beliefs, or sexuality, often while you remain completely unaware of the scope of the tracking [11].

Because high-frequency sounds (above 18 kHz) do not travel far and are entirely invisible to the human ear, ultrasonic cross-device tracking (uXDT) and media tracking have emerged as popular tools. This involves hiding tracking identifiers inside ultrasonic sounds and picking them up using a smartphone's microphone [13].

B. Acoustic Covert Communication

Covert communication happens when devices silently exchange information without the user noticing anything out of the ordinary. While utilizing microphones and speakers on Android devices for this purpose can enable convenient, personalized features—such as cross-device authentication or targeted advertising—it also introduces severe privacy risks when exploited maliciously [13].

A prime example of this threat is how attackers can unmask users who are trying to stay hidden on anonymous networks. Normally, these networks encrypt the source and destination of a message layer by layer to protect the sender's identity.

However, an attacker can bypass this encryption and de-anonymize the user through covert acoustic communication using the following steps [13]:

- 1) **Embedding the Beacon:** The attacker hides high-frequency signals (audio beacons) inside standard audio or video files. They then upload these manipulated media files to a webpage hosted on the anonymous network.
- 2) **User Access:** A target user uses their personal computer to access the anonymous network.
- 3) **Media Playback:** While browsing the compromised webpage, the user clicks play on the synthesized audio or video file.
- 4) **Silent Detection:** Because the hidden beacons are designed to be inaudible, the user hears only normal sound and suspects nothing. However, their device's microphone actively picks up the hidden acoustic signal.
- 5) **Data Extraction:** If the user has a nearby device (like a smartphone) running an app or SDK designed to decode these specific hidden beacons, the microphone's detection triggers the software. The app immediately begins silently harvesting the user's personal data and transmitting it back to the attacker over the internet.

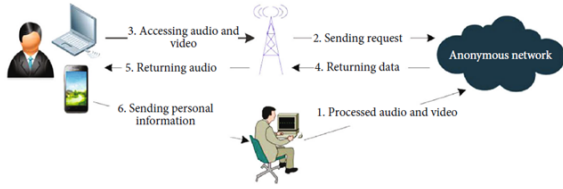


Fig. 4. De-anonymization of anonymous network users using acoustic covert communication [13].

While most security research has focused on complex, AI-driven attacks meant to confuse smart assistants like Google Home or Apple's Siri, a much simpler threat has been overlooked: attackers can use everyday microphones and speakers to broadcast malicious, hidden audio beacons, putting user privacy at serious risk [13].

Research [13] has shown that Android devices can secretly communicate using high-frequency sounds (above 18 kHz) that are completely inaudible to humans. These silent signals are blended into regular audio files, like music or advertisements. When the audio plays, specialized apps on a nearby target device can hear and decode these hidden messages. Several companies already utilize this technology—Shopkick, Lisnr, and Signal360 use ultrasonic beacons in physical stores to send location-based ads to mobile phones, while Silverpush developed a way to hide these tracking codes directly inside TV broadcasts [11].

1) *Case Study: Media Tracking:* Silverpush, an Indian company, created a software development kit (SDK) that other developers could easily integrate into their own shopping apps [13]. This SDK operates quietly in the background: it constantly listens for specific ultrasonic frequencies, and when it detects one, it gathers the user's personal data and reports it straight back to Silverpush's servers. This allowed

the company to secretly collect massive amounts of private information and map out users' daily trajectories.

The Federal Trade Commission (FTC) eventually intervened, sending warning letters to developers and demanding they explicitly tell users if they were using Silverpush's tracking tools. Despite this, researchers still found 234 Android apps—some boasting millions of users—quietly listening for TV beacons in the background. To fight back, modern operating systems tightened their security. Starting with Android 6, apps are required to ask for explicit, run-time permission to use your microphone. iOS took it a step further by displaying a visible warning when an app uses the mic in the background, making it much harder for these tracking apps to operate without raising suspicion [13].

C. Tracking scenarios

Because getting microphone access is now heavily restricted, trackers have found a clever loophole: using the microelectromechanical (MEMS) gyroscopes built into modern smartphones and smartwatches. It turns out that these physical sensors actually vibrate when exposed to specific ultrasonic frequencies (between 19 and 29 kHz). The biggest danger here is that, unlike microphones, apps and websites do not need to ask for any special user permissions to access gyroscope data. This makes the tracking completely stealthy [13], [14].

Here is how this invisible tracking plays out in real-world scenarios:

- 1) **Web Tracking:** You visit a webpage on your laptop, and the site secretly plays an ultrasonic sound through the computer's standard internal speakers. Your phone, sitting nearby on the same desk (within 10 to 50 cm), feels the vibration through its gyroscope. The attacker has now successfully linked your anonymous web browsing session directly to your personal mobile device. Because this can be executed via a simple webpage, the tracking code can easily spread across thousands of sites. [13], [14]
- 2) **TV Tracking:** A television network hides ultrasonic tracking codes inside its broadcast audio. Your phone's gyroscope picks up these silent vibrations from across the room, allowing advertisers to track exactly what shows you are watching and when [13], [14].
- 3) **Location Tracking:** An attacker places high-quality ultrasonic speakers inside a physical location, like a retail store. As you walk within 1 to 3 meters of the speaker, your phone's gyroscope logs the signal, revealing your exact physical location [13], [14].

D. Defense Approaches

To combat these invisible acoustic threats, researchers have proposed a few key defensive strategies:

Speaker Usage Notifications: Devices could add a visual warning—like a specific icon on the screen or a flashing LED light—that turns on whenever the device's loudspeaker is actively broadcasting. If you see the warning but hear absolutely nothing, you immediately know an app or website is trying to secretly broadcast a tracking signal [14].

High-Frequency Audio Filtering: Another robust defense is an OS-level security model called the Ultra-Filter. This works by actively filtering out and blocking high-frequency acoustic signals (specifically from 18 kHz to 20 kHz) before the system can even process them. Researchers recommend integrating this directly into the Android operating system and creating a brand new system permission. This way, if an app genuinely needs to use the ultrasonic sound spectrum, it is forced to explicitly ask the user for permission first, giving control back to the device owner [13].

Auditing App Permissions: Regularly checking which installed apps have microphone access and removing suspicious or unnecessary ones, particularly those using advertising frameworks.

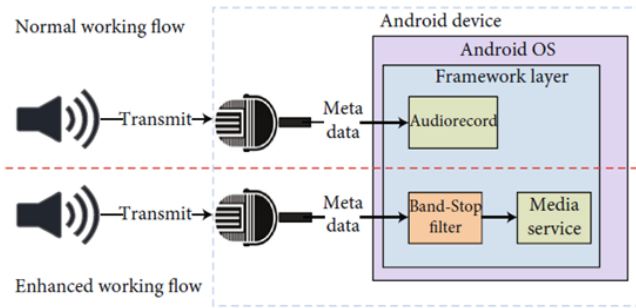


Fig. 5. Security framework based on acoustic high-frequency filtering. The upper half shows the normal working flow; the lower half shows the enhanced flow with the Band-Stop filter [13].

IV. CONCLUSION

Zero-click attacks and acoustic covert communication share one defining quality: they require nothing from the victim. No click, no download, no permission — yet full compromise or persistent tracking is achieved. Real-world cases such as Pegasus, FORCEDENTRY, and Silverpush confirm that these are not theoretical risks but actively deployed techniques that affect millions of ordinary people daily.

This matters beyond the technical. When devices are silently exploited, people lose control over their own information, their behaviour, and ultimately their freedom. Defending against these threats is therefore as much about preserving personal dignity as it is about cybersecurity.

Effective defence requires moving beyond user awareness — because awareness cannot protect against attacks the user never sees. Zero-Trust Architecture, OS-level audio filtering, and continuous device monitoring are the necessary response. As silent threats continue to expand into IoT and beyond, security must be built into systems by design, not left to the individual to maintain.

REFERENCES

[1] “Review of zero-click attacks and zero-trust security model: concepts, architecture, state of the art, and future directions,” *Discover Networks, Springer Nature*, Apr. 2026. [Online]. Available: <https://link.springer.com/article/10.1007/s44354-026-00023-4>

[2] A. A. Younis, Z. Daher, B. Martin, and C. Morgan, “Zero-Click Attacks,” School of Computing and Analytics, Northern Kentucky University, Highland Heights, USA. [Online]. Available: <https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=10216399>

[3] “Developing a Conceptual Framework for Enhanced Cybersecurity Defense Against Zero-Click Exploit,” in *Proc. Springer Nature*, 2026. [Online]. Available: https://link.springer.com/chapter/10.1007/978-3-031-95652-2_21

[4] National Institute of Standards and Technology, “Guidelines for Managing the Security of Mobile Devices in the Enterprise,” *NIST Special Publication 800-124r2*. [Online]. Available: <https://nvlpubs.nist.gov/nistpubs/SpecialPublications/NIST.SP.800-124r2.pdf>

[5] “What Are Zero Click Attacks, and How Can You Protect Against Them?” *Packetlabs*. [Online]. Available: <https://www.packetlabs.net/posts/what-are-zero-click-attacks-and-how-can-you-protect-against-them/>

[6] HP Wolf Security, “Endpoint Security Solutions for Business and Home Computers and Printers,” *HP United Kingdom*. [Online]. Available: <https://www.hp.com/gb-en/security/endpoint-security-solutions.html>

[7] “Why Zero-Click Exploits are the Latest Threat to Executives and HNWIs,” *BlackCloak*. [Online]. Available: <https://blackcloak.io/why-zero-click-exploits-are-the-latest-threat-to-executives-and-hnwis/>

[8] S. Musil, “WhatsApp flaw let attackers install spyware with a phone call,” *CNET*, 13 May 2019. [Online]. Available: <https://www.cnet.com/news/privacy/whatsapp-vulnerability-reportedly-allowed-spyware-attack/>

[9] R. Wood, “Zero-Click Cyber Attacks & AI Agents,” *TimeTrex*, 6 Oct. 2025. [Online]. Available: <https://www.timetrex.com/blog/zero-click-cyber-attacks>

[10] Tahir, “What is a Zero Click Attack? How zero-click attacks hack devices without any user interaction,” *Medium*, Oct. 2025. [Online]. Available: <https://medium.com/@tahirbalarabe2/what-is-a-zero-click-attack-d7f17ec4baca>

[11] N. Roy, H. Hassanieh, and R. R. Choudhury, “BackDoor: Making Microphones Hear Inaudible Sounds,” *IEEE/ACM Trans. Netw.*, vol. 26, no. 4, pp. 1598–1611, Aug. 2018, doi: 10.1109/TNET.2018.2838004. [Online]. Available: <https://ieeexplore.ieee.org/document/8383887>

[12] R. Díaz-Morales, “Cross-Device Tracking: Matching Devices and Cookies,” *arXiv preprint arXiv:1510.01175*, Oct. 2015. [Online]. Available: <https://arxiv.org/pdf/1510.01175>

[13] L. Duan, K. Zhang, B. Cheng, and B. Ren, “Privacy Threats of Acoustic Covert Communication among Smart Mobile Devices,” *Wireless Commun. Mobile Comput.*, vol. 2021, Art. no. 9179100, Jul. 2021, doi: 10.1155/2021/9179100. [Online]. Available: <https://onlinelibrary.wiley.com/doi/10.1155/2021/9179100>

[14] Z. Zhou, W. Diao, X. Liu, and K. Zhang, “Acoustic Fingerprinting Revisited: Generate Stable Device ID Stealthily with Inaudible Sound,” in *Proc. 2014 ACM SIGSAC Conf. Comput. Commun. Security (CCS ’14)*, Scottsdale, AZ, USA, Nov. 2014, pp. 429–440, doi: 10.1145/2660267.2660300. [Online]. Available: <https://dl.acm.org/doi/epdf/10.1145/2660267.2660300>